

Pyrolane — turquoise hydrogen from methane pyrolysis

The Problem

Industry consumes roughly 95 million tonnes of hydrogen a year, and 96% of it is made by steam methane reforming, which releases 9 to 12 tonnes of CO₂ per tonne of hydrogen. Green hydrogen from electrolysis emits nothing but needs 50-55 kWh per kilogram, which makes it two to four times more expensive than the grey hydrogen it is meant to replace.

Refiners, ammonia producers and steelmakers now face binding emissions targets on hydrogen they cannot afford to decarbonise.

Our Solution

Pyrolane splits methane into hydrogen and solid carbon. No CO₂ is produced at any point: the carbon leaves the reactor as a powder rather than a gas.

The reaction needs about 14 kWh per kilogram of hydrogen, roughly a quarter of what electrolysis requires, because breaking a carbon-hydrogen bond costs far less energy than splitting water. Our catalyst holds activity past 900 hours without regeneration, where published systems degrade within 200.

The Market

Industrial hydrogen is a €140B market worldwide. Our entry segment is European refining and ammonia, about €19B a year, where emissions penalties already exceed €80 per tonne of CO₂ and rise on a published schedule through 2030.

We are not competing on green credentials alone. At current European gas prices our delivered cost is below grey hydrogen before any carbon accounting.

Competition

Electrolysis players dominate the decarbonisation narrative and will keep the segments where cheap renewable power is abundant. Three other pyrolysis companies exist; two use molten metal reactors that struggle with carbon separation, and one uses plasma, which reintroduces the electricity cost we are avoiding.

Our catalyst formulation and the continuous carbon extraction mechanism are covered by four granted patents and two pending.

The Team

Ines Marchetti led process engineering at a top-three industrial gas group for twelve years, including three years running a reformer producing 400 tonnes of hydrogen a day. She has commissioned two plants from first drawing to steady state.

Anders Lund spent eleven years on catalyst design, first academically and then at a refining technology licensor, and holds three patents on carbon deposition control — the exact failure mode that kills most pyrolysis reactors.

They are twenty-two people, fourteen of them engineers.

Traction

- pilot reactor running continuously since March, 1,400 hours cumulative
- 62 kg/day of hydrogen at 99.7% purity
- two non-binding offtake agreements with European ammonia producers, 3,000 tonnes/year combined from 2028
- one carbon black offtake signed at €0.9/kg
- €4.2M in non-dilutive innovation grants secured

We are pre-revenue on hydrogen. Nothing is sold at industrial scale yet, and the first commercial unit is scheduled for 2027.

Business Model

Two revenue lines. Hydrogen sold under long-term offtake at a fixed price indexed to gas. Solid carbon sold as carbon black feedstock, currently €0.9/kg, which covers about 30% of our production cost — this is what makes the economics work rather than a subsidy.

Later, licensing the reactor design to industrial operators who prefer to own their own capacity.

Go-To-Market

Long cycles, few buyers. Our first four targets represent 40% of European ammonia output. Sales are technical and run 12 to 18 months, which is why offtake letters precede plant construction.

Funding

We are raising €18M to build the first commercial unit at 1.5 tonnes/day, complete catalyst qualification, and convert the two offtake letters into binding contracts.